



AGRIPRICE:

Tracking Agricultural Prices Amid Typhoons

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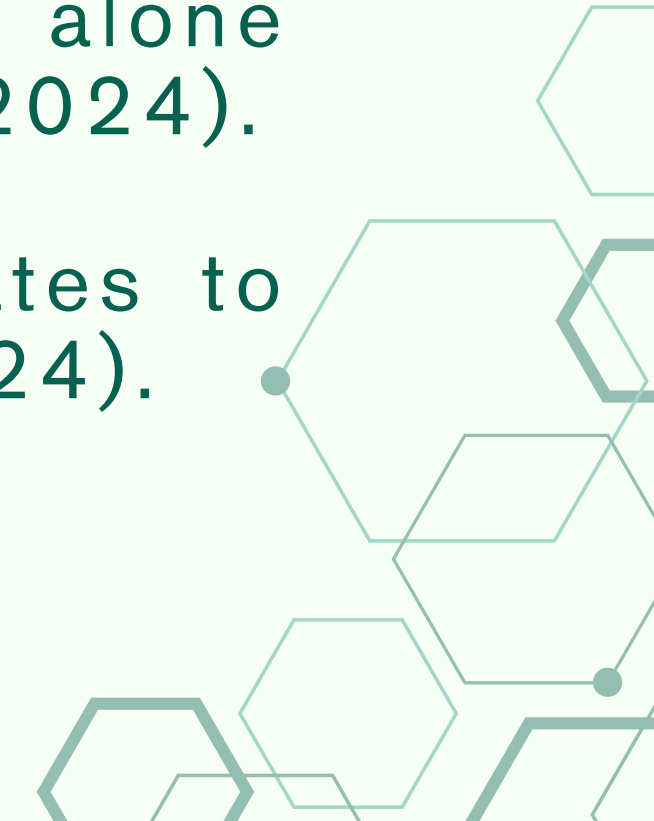


CLIMATE CHANGE IS NOT JUST AN ENVIRONMENTAL PROBLEM; IT'S A DIRECT ECONOMIC THREAT TO THE COST OF FOOD.

Global Frontline: The country experiences ~ approx 20 tropical cyclones annually (PAGASA, n.d.).

Economic Blow: Typhoons Kristine and Leon (2024) alone resulted in P6.83 Billion in agricultural losses (Inquirer.net, 2024).

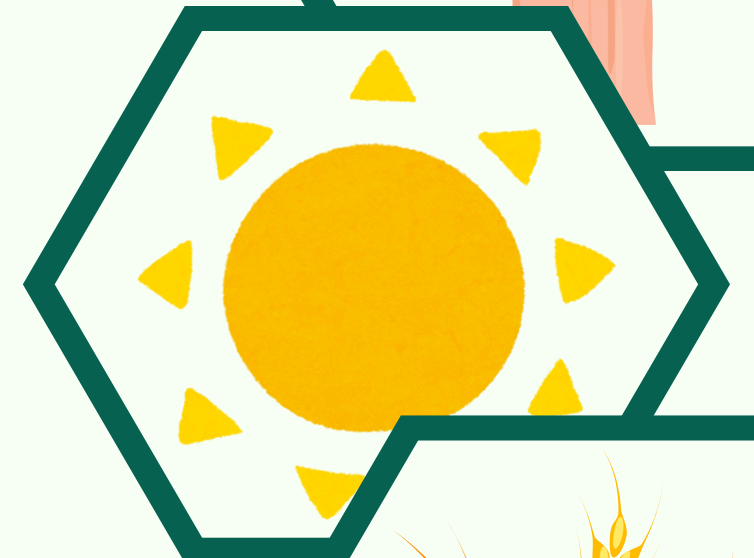
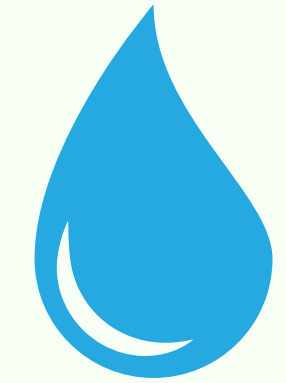
Consumer Burden: This supply shock immediately translates to higher food prices for vulnerable households (PCAARRD, 2024).



THE MARKET GAP: ANTICIPATION VS. REACTION

EXISTING MONITORING FOCUSES ON DAMAGE ASSESSMENT AFTER THE EVENT, BUT WE LACK THE TOOLS FOR:

1. **Anticipation:** Predicting which commodity will spike first.
2. **Granularity:** Knowing which regions are structurally most prone to price chaos.
3. **Proactive Policy:** Developing evidence-based intervention strategies. Our Goal: Move from post-disaster reaction to proactive, data-informed anticipation.

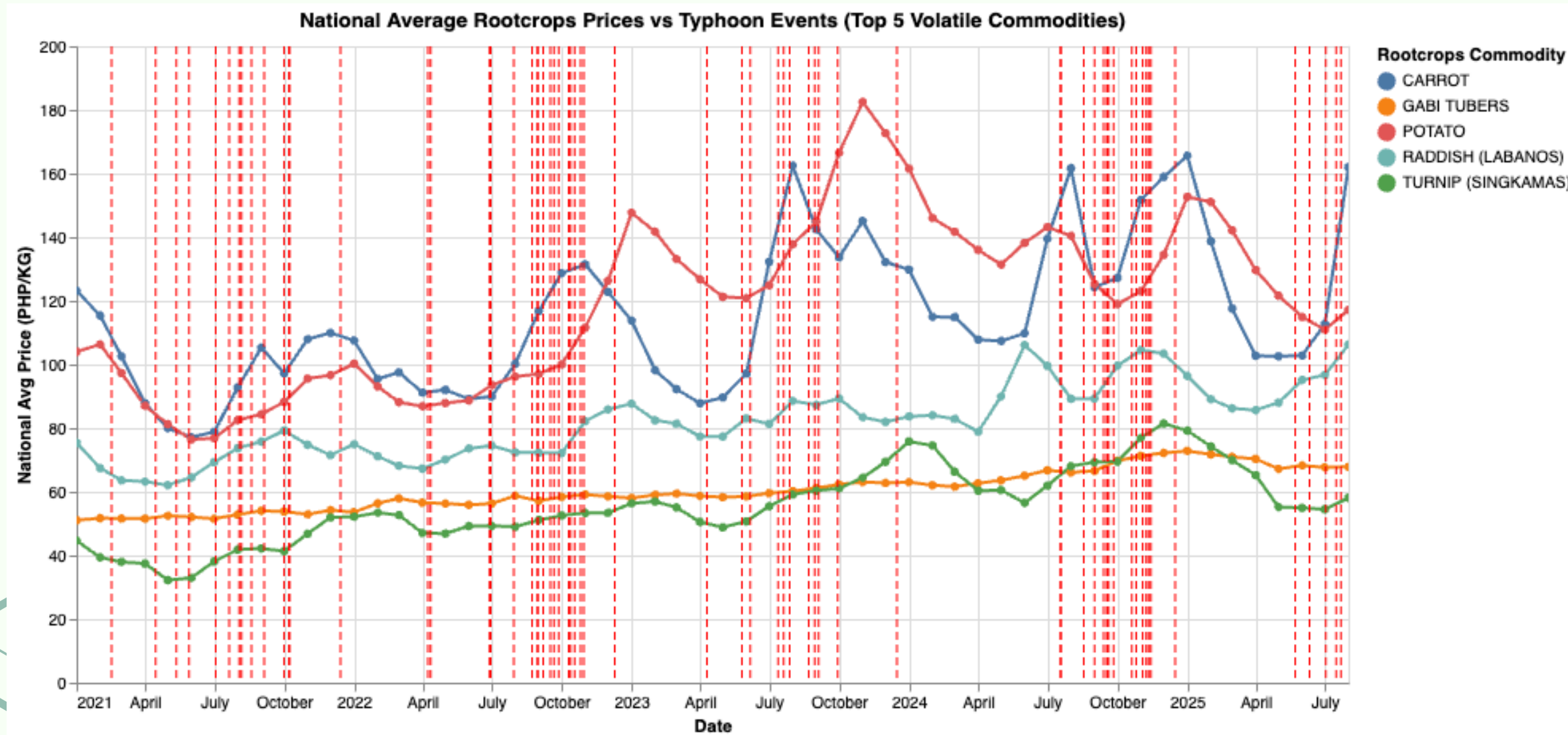


DATA INTEGRATION: COMBINING TYPHOONS AND PRICES

- **Climate Data (PAGASA):** Historical tropical cyclones – timing and intensity.
- **Sources:** PAGASA Annual Reports ([link](#)) and PAGASA Preliminary Reports ([link](#))
- **Price Data (PSA):** Monthly retail prices of selected commodities.
- **Source:** PSA OpenSTAT ([link](#))



NATIONAL AVERAGE ROOTCROPS PRICES VS TYPHOON EVENTS (TOP 5 VOLATILE COMMODITIES)



Typhoon Events Clearly Drive Root Crop Price Spikes

- **Direct Impact:** Typhoons consistently precede major price spikes for root crops, especially Potato and Carrot.
- **Delayed Effect:** The visual confirms the lag: prices often rise in the months following a cluster of typhoons.
- **Key Insight:** Typhoons are a major, consistent shock that directly causes high price volatility for these key root crops. (Other crop types are analyzed separately.)

THE AGRIPRICE RISK METRICS

Metric 1: Price Spike Frequency (% of Spike Days)

- **Definition:** How often does the price exceed Median + 1.5 x IQR?
- **Significance:** Measures acute, short-term shocks.

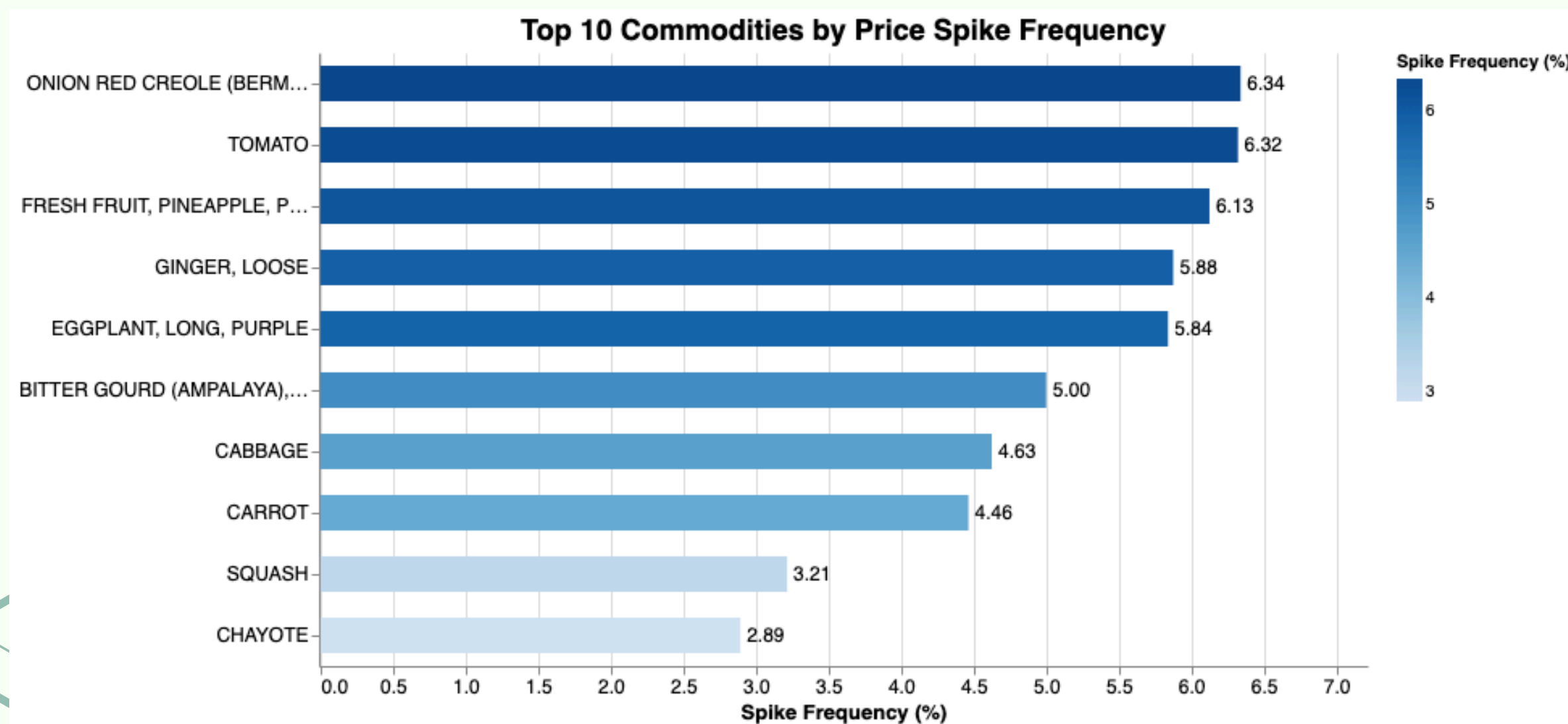
Metric 2: Price Volatility (Standard Deviation)

- **Definition:** The dispersion of retail prices over time.
- **Significance:** Measures chronic, long-term instability.

Metric 3: Price Lag (Mean Months to First Spike)

- **Definition:** Time elapsed between a Typhoon's entry into the PAR and the commodity's first price spike.
- **Significance:** Measures market speed and vulnerability to immediate supply disruption.

FINDING 1: WHICH COMMODITIES SPIKE MOST FREQUENTLY?



Red Onion Leads Price Spike Frequency

- **Red Onion** is the most volatile commodity at 6.34%.
- **Top 4 Cluster:** Tomato (6.32%), Pineapple (6.13%), and Ginger (5.88%) are nearly as volatile.
- **The Range:** Frequency spans **2.89% to 6.34%**, highlighting market sensitivity across all key produce.

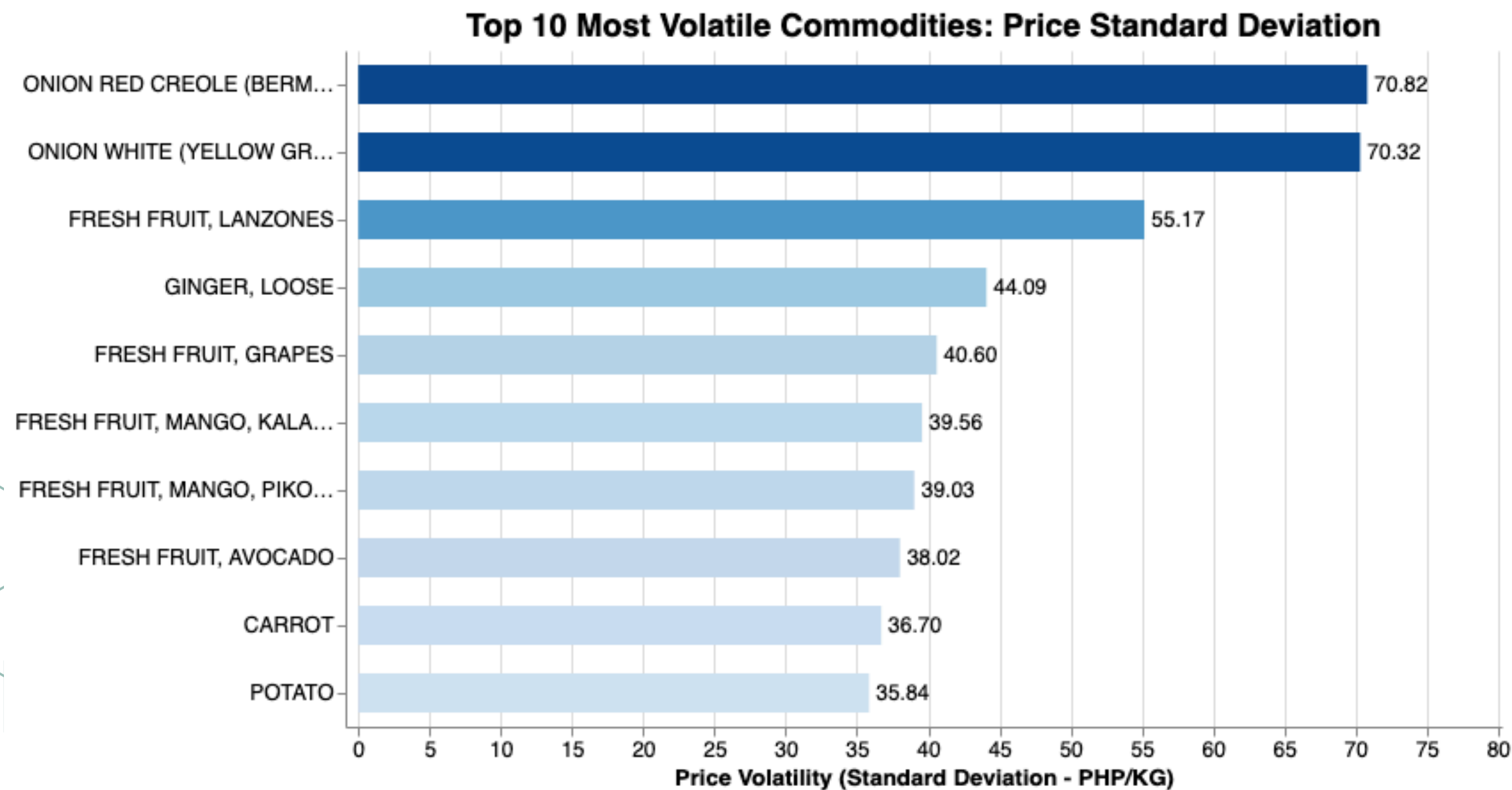
FINDING 2: REGIONAL CONCENTRATION OF PRICE SPIKES

	Region	Total_Spikes
4	III - Central Luzon	716
7	MIMAROPA Region	539
5	IV-A - CALABARZON	472
9	V - Bicol Region	412
3	II - Cagayan Valley	310

- **Highest Spikes:** Region III - Central Luzon leads significantly with 716 total spikes.
- **Top 3:** The concentration of price spikes is in Central Luzon, MIMAROPA, and CALABARZON.
- **Action:** Interventions must be focused on these 3 regions to address the highest frequency of price irregularities.

Red Onion Leads Price Spike Frequency

FINDING 3: WHICH COMMODITIES SHOW THE HIGHEST PRICE VOLATILITY?



Onions Dominate Volatility in Price Swings

- **Onion Volatility:** Both Red and White Onions lead by a huge margin, with price swing intensity (standard deviation) over 70 PHP/KG.
- **The Gap:** The third-ranked item, Lanzones (55.17), is significantly less volatile, marking a sharp drop-off.
- **Key Takeaway:** Onion prices are not only spiking frequently but the size of those price changes is substantially higher than any other item.

FINDING 4: HOW QUICKLY DO PRICES REACT AFTER A TYPHOON EVENT?

TOP 5

	Commodity_Name	mean	median	count
21	GARLIC, LOCAL	0.900000	1.0	10
13	FRESH FRUIT, GUYABANO	0.956522	1.0	23
19	FRESH FRUIT, PINEAPPLE, PINYA, MEDIUM	0.969697	1.0	66
14	FRESH FRUIT, LANZONES	1.000000	1.0	19
32	SQUASH	1.000000	1.0	61

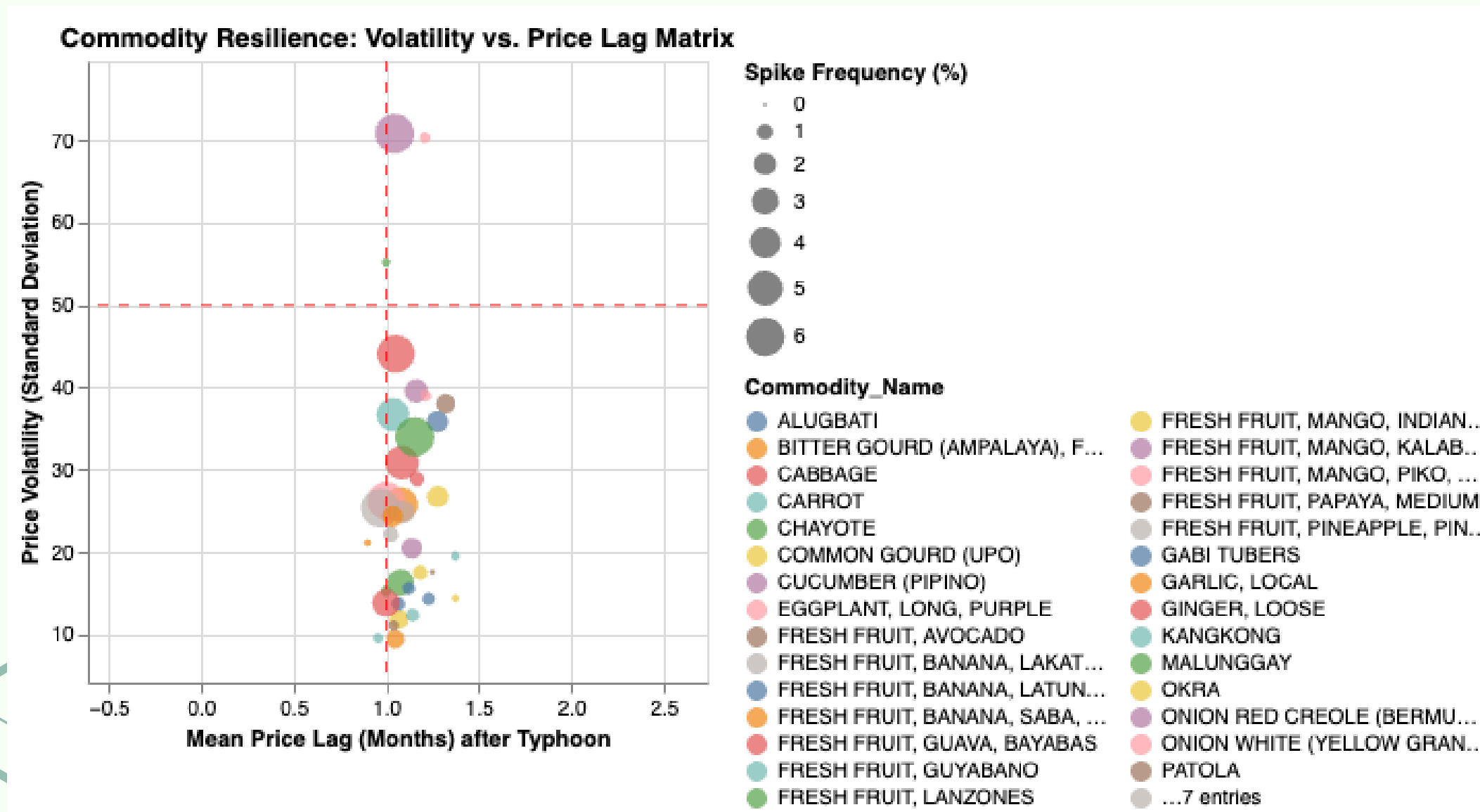
BOTTOM 5

	Commodity_Name	mean	median	count
30	POTATO	1.277778	1.0	18
25	OKRA	1.279070	1.0	43
8	FRESH FRUIT, AVOCADO	1.322581	1.0	31
23	KANGKONG	1.375000	1.0	8
35	TURNIP (SINGKAMAS)	1.375000	1.0	16

Price Spikes Consistently Materialize Within One Month

- Spikes Consistently Hit Within **One Month**
- **Median Lag:** 1.0 month for all commodities.
- **Response Time:** Price spikes materialize in a narrow window (0.9 to 1.375 months).
- **Extremes:** Garlic is fastest (0.9 months); Turnip/Kangkong show the maximum delay (1.375 months).
- **Key Insight:** Price spikes happen rapidly and are not delayed by multiple months.

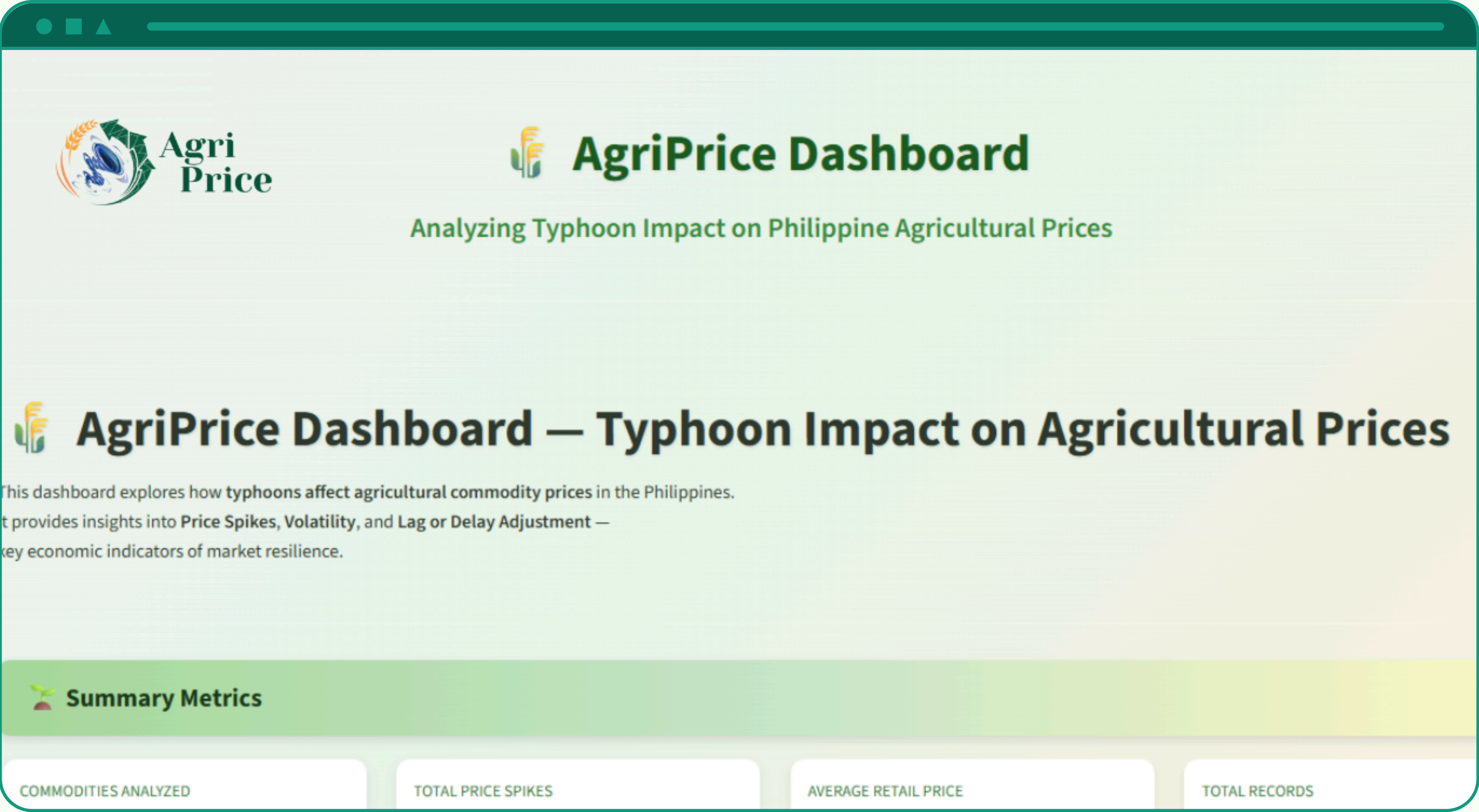
FINDING 5: COMMODITY RESILIENCE ANALYSIS



Market Response Varies Across Three Resilience Groups

- **Least Resilient:** Red and White Onions are outliers, combining a slow spike response (around 1 month) with extreme subsequent volatility.
- **Most Resilient:** Carrot and Ginger react fast to typhoons, showing a fast spike (<1 month).
- **Majority Trend:** Most commodities show a delayed spike (around 1 month) but achieve low subsequent volatility, meaning the market shock is slow but manageable.

AGRIPRICE: OUR INTERACTIVE DASHBOARD



Product Overview: A unified, easy-to-use web application for all stakeholders.

Purpose: Visualize market stability and climate risk in one interface.

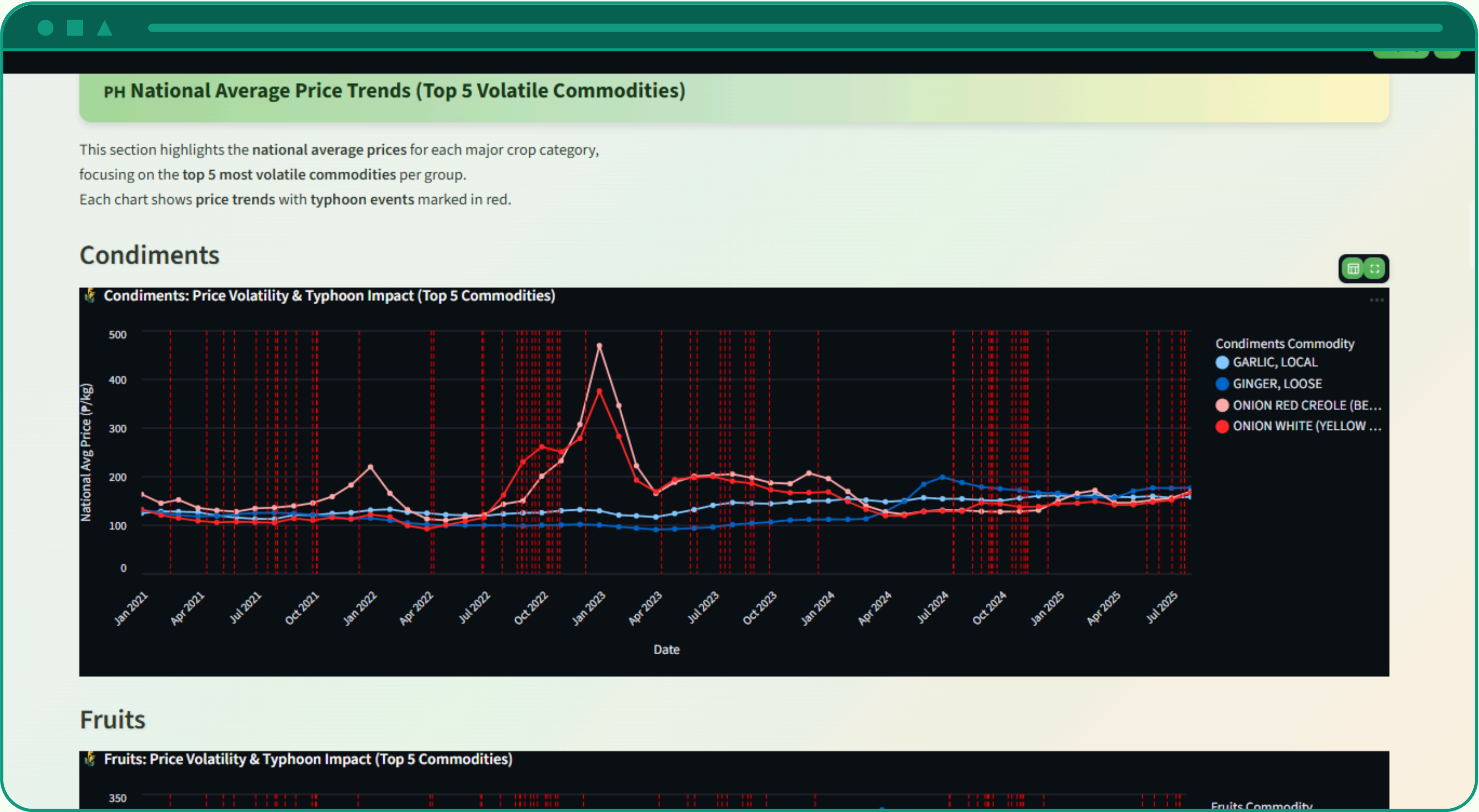
Users: Policy analysts (DA/DTI), Agricultural Cooperatives, and Consumers.

FEATURE 1: VISUALIZING PRICE SPIKES ON A TIMELINE

- *Demo Point:* Easily show the correlation: "Notice how Cabbage's price line immediately jumps after every red typhoon line in the Jul-Oct season."
- *Value:* **Immediate visual validation** of climate-price correlation.



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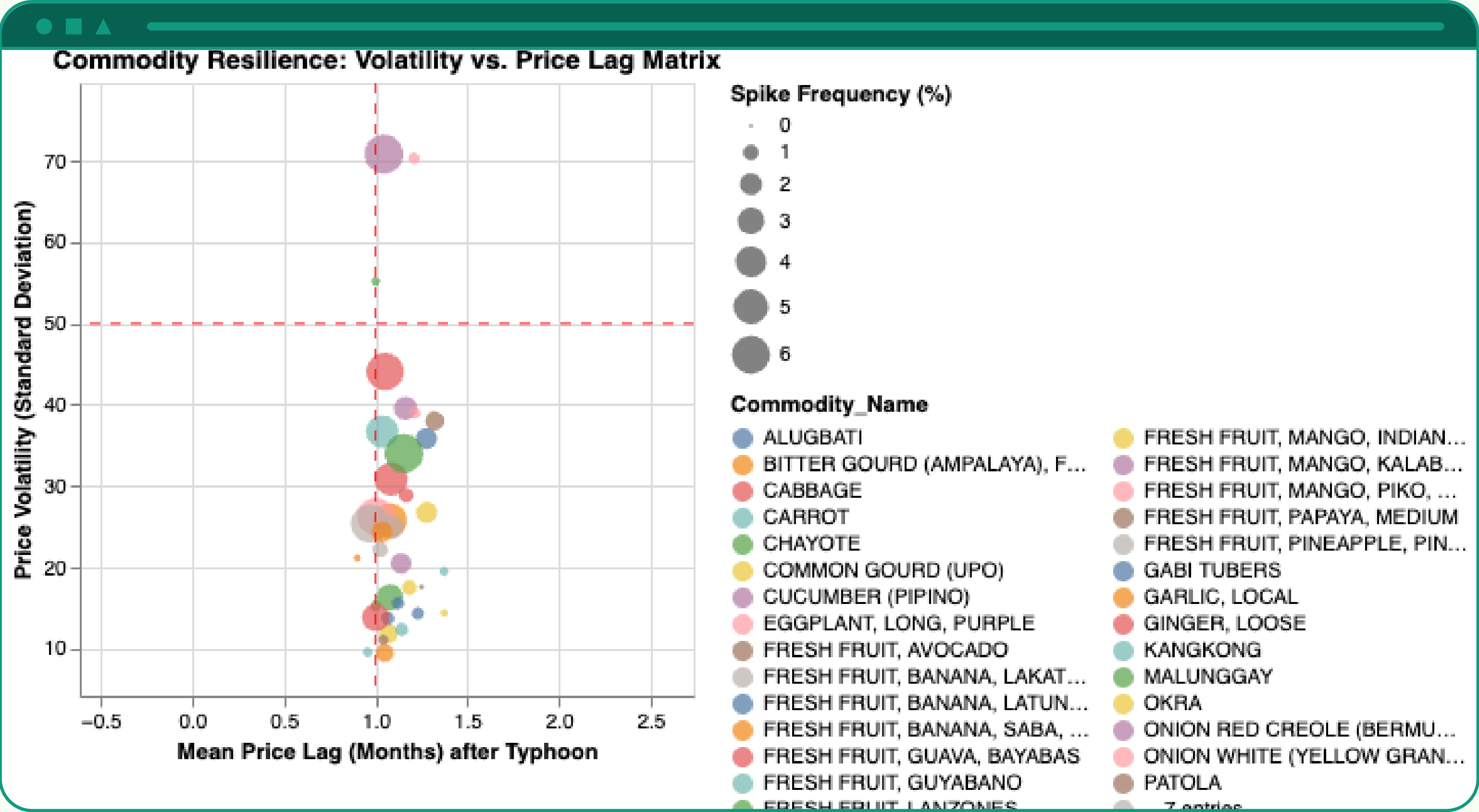
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FEATURE 2: THE RESILIENCE MATRIX (LAG VS. VOLATILITY SCATTER)

- *Demo Point:* Hover over a commodity (e.g., Tomato) to show its high volatility (Y-axis) and fast reaction speed (X-axis).
- *Value:* **Data-driven prioritization**—pinpointing exactly which commodity needs intervention based on its measured risk profile.



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CONCLUSION

AgriPrice moves the Philippines beyond reactive disaster response towards a proactive, data-informed strategy for climate resilience and food security.



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